
实验 2：数据定义与数据操纵（实验报告）

（2026 春季，智能软件与工程学院）

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（说明：实验报告请使用 PDF 格式提交，截止日期：2026.05.12）

实验环境

[如果你不是使用本课程建议的 MySQL 数据库，请用一句话介绍你使用的操作系统、数据库管理系统、软件版本]

实验过程

[所有要求提交的 **SQL 语句文本**和**执行结果截图**]

第一部分：数据库创建

```
use mysql;
drop schema if exists myschool26;
create schema myschool26 charset gb2312 collate gb2312_bin;
use myschool26;
```

```
create table students(
    sno char(9) not null,
    sname char(20) not null,
    ssex char(2) not null,
    birthday date not null,
    dept char(20),
    enrolldate date,
    primary key (sno),
    check (ssex in ('男', '女'))
);
```

```
create table teachers(
    tno char(9) not null,
    tname char(20) not null,
    dept char(20),
    primary key (tno)
);
```

```
create table courses(  
    cno char(8) not null,  
    cname char(40) not null,  
    credit int not null,  
    chours int not null,  
    dept char(20),  
    primary key (cno),  
    check (credit > 0 and credit < 10)  
);  
  
create table courseclass(  
    clsno char(8) not null,  
    cno char(8) not null,  
    tno char(9),  
    clsyear int not null,  
    clsterm char(4) not null,  
    primary key (clsno),  
    check (clsterm in ('春季', '秋季', '暑期')),  
    foreign key (cno) references courses(cno)  
        on update cascade  
        on delete restrict,  
    foreign key (tno) references teachers(tno)  
        on update cascade  
        on delete set null  
);  
  
create table enrollment(  
    sno char(9) not null,  
    clsno char(8) not null,  
    grade int,  
    primary key (sno, clsno),  
    foreign key (sno) references students(sno)  
        on update cascade  
        on delete restrict,  
    foreign key (clsno) references courseclass(clsno)  
        on update cascade  
        on delete restrict,  
    check (grade >= 0 and grade <= 100)  
);
```

第二部分：初始数据加载（不需要提交）

第三部分：数据查询

1.

```
select cno, cname, dept
from courses
where cname like '%数%' -- 包含“数”字的课程
order by dept asc, cno asc;
```

```
77  /* 第二部分：初始数据加载（不需要提交） */
78  /* 第三部分：数据查询 */
79  • select cno, cname, dept
80    from courses
81   where cname like '%数%' -- 包含“数”字的课程
82   order by dept asc, cno asc;
```

	cno	cname	dept
▶	1201	高等数学(I)	计算机
	2205	数据结构	计算机
	3208	数据库概论	计算机
	3308	数据库技术	人工智能
	1101	高等数学	数学
	2105	数据结构	数学
	2106	数学分析	数学
*	NULL	NULL	NULL

2.

```
select sno, sname, dept
from students
where sname like '_静%' -- 名字第二个字是“静”的学生， '_'表示任意一个字符（不是字节）
-- 虽然中文是2个字节，但是_匹配一个字符。%匹配任意长度的字符串（包括空字符串）
order by sname asc;
```

```
84  • select sno, sname, dept
85    from students
86   where sname like '_静%' -- 名字第二个字是“静”的学生， '_'表示任意一个字符（不是字节）
87   -- 虽然中文是2个字节，但是_匹配一个字符。%匹配任意长度的字符串（包括空字符串）
88   order by sname asc;
```

	sno	sname	dept
▶	221250159	陈静颖	人工智能
	211210091	郭静	计算机
	241210019	王静琳	计算机
*	NULL	NULL	NULL

3.

```
select s.sno, s.sname, c.cno, c.cname
from enrollment e
```

join students s on e.sno = s.sno -- 必须要 join, 因为 sname 这些值在 enrollment 表里没有, 必须要连接到 students 表才能获取

-- on 后面是连接条件, 表示 enrollment 表的 sno 和 students 表的 sno 相等时才连接

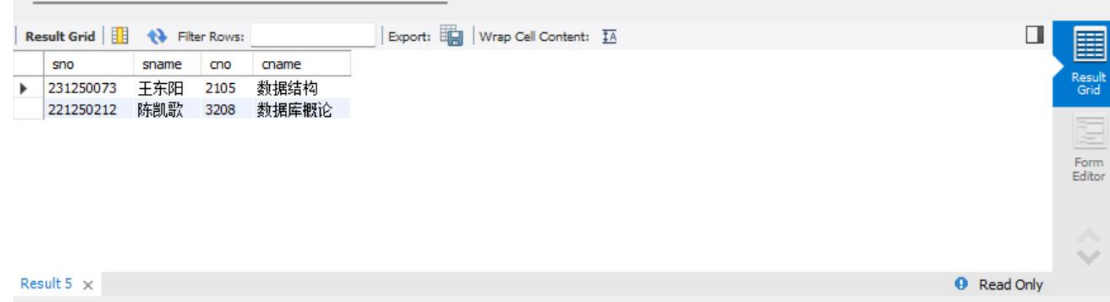
join courseclass cc on e.clsno = cc.clsno

join courses c on cc.cno = c.cno

where e.grade is null

order by c.cno asc, s.sno asc;

```
90 • select s.sno, s.sname, c.cno, c.cname
91 from enrollment e
92 join students s on e.sno = s.sno -- 必须要join, 因为sname这些值在enrollment表里没有, 必须要连接到students表才能获
93 -- on后面是连接条件, 表示enrollment表的sno和students表的sno相等时才连接
94 join courseclass cc on e.clsno = cc.clsno
95 join courses c on cc.cno = c.cno
96 where e.grade is null
97 order by c.cno asc, s.sno asc;
```



sno	sname	cno	cname
231250073	王东阳	2105	数据结构
221250212	陈凯歌	3208	数据库概论

4.

select s.sno, s.sname,

count(*) as course_count, -- 统计每个学生选了多少门课程, 直接用*就行, 因为每一行代表选了一门课了

sum(c.credit) as total_credits, -- 统计每个学生选了多少学分, sum 函数会自动忽略 null 值, 所以不需要在 where 里加 grade is not null 的条件了

avg(e.grade) as avg_grade

from enrollment e

join students s on e.sno = s.sno

join courseclass cc on e.clsno = cc.clsno

join courses c on cc.cno = c.cno

where year(s.enrolldate) = 2023

and e.grade is not null

group by s.sno

having min(e.grade) >= 60 -- 这里不能放到 where 里, 因为 where 是在分组前过滤数据的, 而 having 是在分组后过滤数据的, 只有分组后才能计算每个学生的最低成绩

-- 聚合函数不能直接在 where 里使用, 必须要在 group by 后面使用 having 来过滤分组后的结果

order by s.sno asc;

```

99 • select s.sno, s.sname,
100        count(*) as course_count, -- 统计每个学生选了多少门课程，直接用*就行，因为每一行代表选了一门课了
101        sum(c.credit) as total_credits, -- 统计每个学生选了多少学分，sum函数会自动忽略null值，所以不需要在where里加
102        avg(e.grade) as avg_grade
103    from enrollment e
104   join students s on e.sno = s.sno
105  join courseclass cc on e.clsno = cc.clsno
106  join courses c on cc.cno = c.cno
107   where year(s.enrolldate) = 2023
108      and e.grade is not null
109   group by s.sno
110  having min(e.grade) >= 60 -- 这里不能放到where里，因为where是在分组前过滤数据的，而having是在分组后过滤数据的，只
111     -- 聚合函数不能直接在where里使用，必须要在group by后面使用having来过滤分组后的结果
112  order by s.sno asc;

```

sno	sname	course_count	total_credits	avg_grade
231220042	李语诚	3	11	83.0000
231250006	吴晓静	4	17	79.5000
231250013	吴川江	3	11	81.0000
231250020	吴雅婷	3	11	77.3333
231250021	吴迪	3	12	90.6667
231250045	蒋权友	4	17	86.2500
231250054	郑韬	3	11	91.6667
231250064	李依娜	3	12	79.0000
231250073	于东明	3	13	83.3333

5.

```

select t.tno, t.tname,
count(*) as courseclass_cnt,
sum(c.chours) as total_hours
from courseclass cc
join teachers t on cc.tno = t.tno
join courses c on cc.cno = c.cno
where cc.tno is not null
group by t.tno
order by total_hours desc, t.tno asc;

```

```

114 • select t.tno, t.tname,
115        count(*) as courseclass_cnt,
116        sum(c.chours) as total_hours
117    from courseclass cc
118   join teachers t on cc.tno = t.tno
119   join courses c on cc.cno = c.cno
120   where cc.tno is not null
121   group by t.tno
122   order by total_hours desc, t.tno asc;
123

```

tno	tname	courseclass_cnt	total_hours
601	王家庚	8	576
602	朱荣华	6	486
802	张无忌	7	432
805	杨玉昆	3	324
704	张家荣	5	236
703	陈坤	4	216
811	李晋城	2	180

6.

```
select left(sname, 1) as surname, count(*) as surname_count -- 用 left 函数获取姓氏, 统计每个姓氏的人数
```

```
from students
```

```
group by surname
```

```
order by surname_count desc, surname asc;
```

```
124 • select left(sname, 1) as surname, count(*) as surname_count -- 用left函数获取姓氏, 统计每个姓氏的人数
125 from students
126 group by surname
127 order by surname_count desc, surname asc;
128
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
surname	surname_count		
李	8		
吴	8		
陈	6		
王	6		
张	5		
刘	3		
蒋	2		
郭	1		
郑	1		

7.

```
select s.dept, s.sno, s.sname
```

```
from students s
```

```
where year(s.enrolldate) = 2021
```

```
and exists (
```

```
    select * from courses c
```

```
    where c.dept = s.dept
```

```
    and not exists ( -- 双重 exists, 表示存在一个本学院开设的课程没及格
```

```
        select * from enrollment e
```

```
        join courseclass cc on cc.clsno = e.clsno
```

```
        where e.sno = s.sno
```

```
        and cc.cno = c.cno
```

```
        and e.grade >= 60
```

```
    )
```

```
)
```

```
order by s.dept asc, s.sno asc;
```

```

129 • select s.dept, s.sno, s.sname
130 from students s
131 where year(s.enrolldate) = 2021
132 and exists (
133     select * from courses c
134     where c.dept = s.dept
135     and not exists ( -- 双重exists, 表示存在一个本学院开设的课程没及格
136         select * from enrollment e
137         join courseclass cc on cc.clsno = e.clsno
138         where e.sno = s.sno
139         and cc.cno = c.cno
140         and e.grade >= 60
141     )
142 )
143 order by s.dept asc, s.sno asc;

```

Result Grid

	dept	sno	sname
▶	计算机	211210043	刘家君
	计算机	211210166	蒋超
	数学	211210088	吴小芳
*	NULL	NULL	NULL

8.

```

select s.dept, s.sno, s.sname
from students s
where year(s.enrolldate) = 2021
and not exists ( -- 就是加一个 not
    select * from courses c
    where c.dept = s.dept
    and not exists (
        select * from enrollment e
        join courseclass cc on cc.clsno = e.clsno
        where e.sno = s.sno
        and cc.cno = c.cno
        and e.grade >= 60
    )
)
order by s.dept asc, s.sno asc;

```

```

145 • select s.dept, s.sno, s.sname
146   from students s
147   where year(s.enrolldate) = 2021
148   and not exists ( -- 就是加一个not
149     select * from courses c
150     where c.dept = s.dept
151     and not exists (
152       select * from enrollment e
153       join courseclass cc on cc.clsno = e.clsno
154       where e.sno = s.sno
155       and cc.cno = c.cno
156       and e.grade >= 60
157     )
158   )
159   order by s.dept asc, s.sno asc;

```

dept	sno	sname
计算机	211210001	陈圣威
计算机	211210091	郭静
计算机	211210119	王伟
人工智能	211210042	刘菲
人工智能	211210112	李杰
数学	211210155	李程锦
NULL	NULL	NULL

9.

select t.tno, t.tname, t.dept,

min(year(cc.clsyear)) as first_year, -- 天然保留 null 值, 表示没有开过课的教师

max(year(cc.clsyear)) as last_year

from teachers t

left join courseclass cc on t.tno = cc.tno -- 如果不用 left join, 则无法统计没有开课的教师, 也就无法保留空值

group by t.tno

order by t.dept asc, t.tno asc;

```

161 • select t.tno, t.tname, t.dept,
162       min(year(cc.clsyear)) as first_year, -- 天然保留null值, 表示没有开过课的教师
163       max(year(cc.clsyear)) as last_year
164   from teachers t
165   left join courseclass cc on t.tno = cc.tno -- 如果不用left join, 则无法统计没有开课的教师, 也就无法保留空值
166   group by t.tno
167   order by t.dept asc, t.tno asc;

```

tno	tname	dept	first_year	last_year
602	朱荣华	计算机	NULL	NULL
704	张家荣	计算机	NULL	NULL
777	李云龙	计算机	NULL	NULL
811	李晋城	计算机	NULL	NULL
703	陈坤	人工智能	NULL	NULL
802	张无忌	人工智能	NULL	NULL
601	王家庚	数学	NULL	NULL
805	杨玉昆	数学	NULL	NULL

10.

select c.cno, c.cname, cc.clsno, cc.clsyear, cc.clsterm

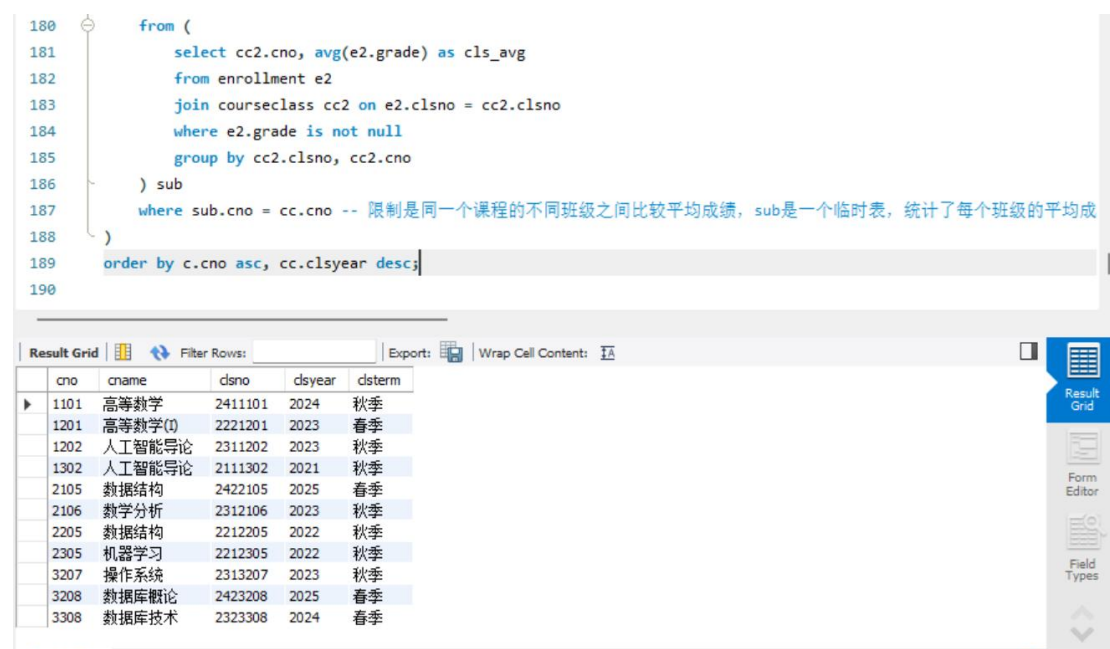
from courseclass cc

join courses c on cc.cno = c.cno


```

join (
    select clsno, avg(grade) as cls_avg
    from enrollment
    where grade is not null
    group by clsno
) ca on ca.clsno = cc.clsno -- classnumber and avarage grade for each class
where ca.cls_avg = (
    select max(sub.cls_avg)
    from (
        select cc2.cno, avg(e2.grade) as cls_avg
        from enrollment e2
        join courseclass cc2 on e2.clsno = cc2.clsno
        where e2.grade is not null
        group by cc2.clsno, cc2.cno
    ) sub
    where sub.cno = cc.cno -- 限制是同一个课程的不同班级之间比较平均成绩，sub 是一个
    临时表，统计了每个班级的平均成绩，然后在外层查询中找出每个课程的最高平均成绩
)
order by c.cno asc, cc.clsyear desc;

```



The screenshot shows a SQL IDE interface. The top part displays a SQL query with line numbers 180 to 190. The query is a complex join and aggregation query. Below the query editor, there is a 'Result Grid' tab. The result grid shows a table with columns: cno, cname, clsno, clsyear, and clsyear. The table contains 15 rows of data, representing different courses and their classes. The right sidebar contains icons for 'Result Grid', 'Form Editor', and 'Field Types'.

cno	cname	clsno	clsyear	clsyear
1101	高等数学	2411101	2024	秋季
1201	高等数学(I)	2221201	2023	春季
1202	人工智能导论	2311202	2023	秋季
1302	人工智能导论	2111302	2021	秋季
2105	数据结构	2422105	2025	春季
2106	数学分析	2312106	2023	秋季
2205	数据结构	2212205	2022	秋季
2305	机器学习	2212305	2022	秋季
3207	操作系统	2313207	2023	秋季
3208	数据库概论	2423208	2025	春季
3308	数据库技术	2323308	2024	春季

第四部分：数据更新

(一) 元组插入

```

set autocommit=0;
insert into courseclass values('2513208','3208', '704', '2025', '秋季');

```

```

insert into enrollment(sno, clsno, grade)
select sno, '2513208', null
from students
where dept = '计算机学院' and year(enrolldate) = 2024;

```

COMMIT;

```

select s.sno, s.dept as student_dept, c.cname, c.dept as course_dept, t.tname, cc.clsyear,
cc.clsterm
from enrollment e
join students s on e.sno = s.sno
join courseclass cc on e.clsno = cc.clsno
join courses c on cc.cno = c.cno
left join teachers t on cc.tno = t.tno
where e.grade is NULL
order by s.sno, c.cno;

```

```

203 • select s.sno, s.dept as student_dept, c.cname, c.dept as course_dept, t.tname, cc.clsyear, cc.clsterm
204   from enrollment e
205   join students s on e.sno = s.sno
206   join courseclass cc on e.clsno = cc.clsno
207   join courses c on cc.cno = c.cno
208   left join teachers t on cc.tno = t.tno
209   where e.grade is NULL
210   order by s.sno, c.cno;
211

```

Result Grid						
Filter Rows:						
Export:						
Wrap Cell Content:						
sno	student_dept	cname	course_dept	tname	clsyear	clsterm
221250212	计算机	数据库概论	计算机	张家荣	2025	春季
231250073	数学	数据结构	数学	朱荣华	2025	春季

(二) 元组修改

```

set autocommit=0;
update enrollment
set grade = 68
where sno = '211210166' and clsno = '2111202';

```

```

select s.sno, s.sname, s.dept
from students s
where year(s.enrolldate) = 2021
and not exists(
    select * from courses c
    where c.dept = '计算机学院'
    and not exists(

```

```

select * from enrollment e
join courseclass cc on e.clsno = cc.clsno
where e.sno = s.sno
and cc.cno = c.cno
and e.grade >= 60
)
)
order by s.sno;

```

ROLLBACK;

```

219   from students s
220   where year(s.enrolldate) = 2021
221   and not exists(
222       select * from courses c
223       where c.dept = '计算机学院'
224   and not exists(
225       select * from enrollment e
226       join courseclass cc on e.clsno = cc.clsno
227       where e.sno = s.sno
228       and cc.cno = c.cno
229       and e.grade >= 60
230   )
231   )

```

Result Grid			Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
sno	sname	dept				
211210001	陈圣威	计算机				
211210042	刘菲	人工智能				
211210043	刘家君	计算机				
211210088	吴小芳	数学				
211210091	郭静	计算机				
211210112	李杰	人工智能				
211210119	王伟	计算机				
211210155	李程锦	数学				
211210166	蒋超	计算机				
NULL	NULL	NULL				

```

update enrollment
join students s on enrollment.sno = s.sno
join courseclass cc on enrollment.clsno = cc.clsno
join courses c on cc.cno = c.cno
set grade = NULL
where c.dept != s.dept;

```

ROLLBACK;

97	15:28:35	ROLLBACK	0 row(s) affected
98	15:28:38	update enrollment join students s on enrollment.sno = s.sno join courseclass cc on enrollment.clsno = cc.clsno join courses c on c...	23 row(s) affected Rows matched: 23 Changed: 23 Warnings: 0
99	15:29:00	ROLLBACK	0 row(s) affected

(三) 元组删除

```

set autocommit=0;
delete from enrollment
where clsno in (
    select clsno from courseclass

```

```

where cno in (
    select cno from courses
    where cname = '人工智能导论' and dept = '计算机学院'
)
);
delete from courseclass
where cno in (
    select cno from courses
    where cname = '人工智能导论' and dept = '计算机学院'
);
delete from courses
where cname = '人工智能导论' and dept = '计算机学院';

ROLLBACK;

```

```

delete from enrollment
where clsno = '2513208';

```

```

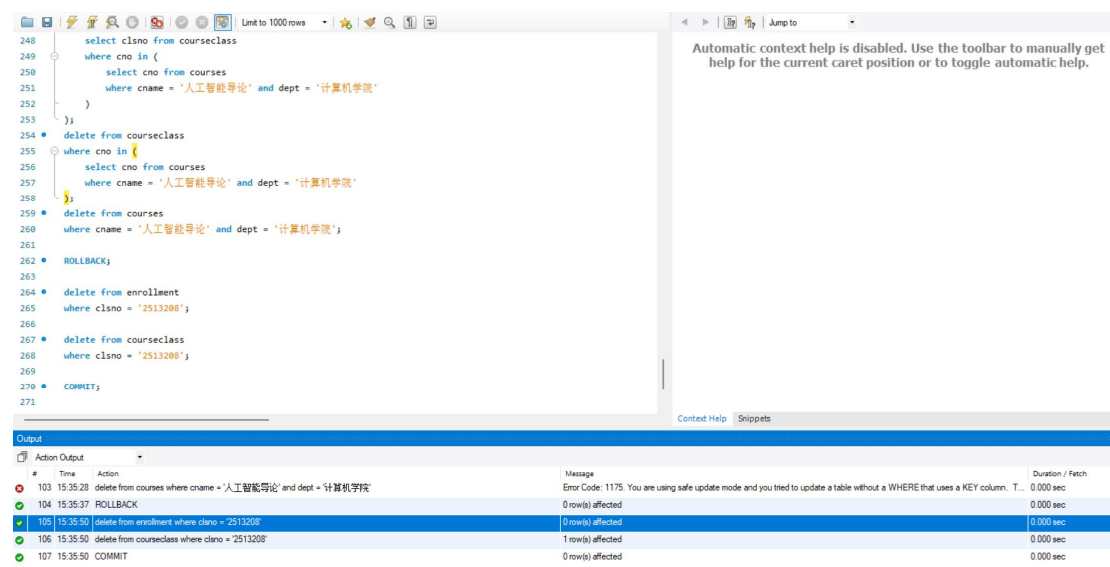
delete from courseclass
where clsno = '2513208';

```

```

COMMIT;

```



The screenshot shows a SQL IDE interface. The main editor displays a SQL script with line numbers 248 to 271. The script includes a subquery to delete from courseclass, followed by deleting from courses, a ROLLBACK, and then deleting from enrollment and courseclass based on a specific class number, and finally a COMMIT. The Output window at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
103	15:35:28	delete from courses where cname = '人工智能导论' and dept = '计算机学院'	Error Code: 1175. You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column. T...	0.000 sec
104	15:35:37	ROLLBACK	0 row(s) affected	0.000 sec
105	15:35:50	delete from enrollment where clsno = '2513208'	1 row(s) affected	0.000 sec
106	15:35:50	delete from courseclass where clsno = '2513208'	1 row(s) affected	0.000 sec
107	15:35:50	COMMIT	0 row(s) affected	0.000 sec

第五部分：视图创建与访问

1.

```

create view avg_course_grade(cno, total_students, avg_grade) as
select cc.cno, count(*) as total_students, avg(e.grade) as avg_grade

```

```

from courseclass cc
left join enrollment e on cc.clsno = e.clsno
where e.grade is not null
group by cc.cno;

```

32 18:56:43 create view avg_course_grade(cno, total_students, avg_grade) as select cc.cno, count(*) as total_students, avg(e.grade) as avg... 0 row(s) affected

2. (1)

```

create view total_course_students(cno, total_students, num_greater_avg_grade) as
select cc.cno,
       count(*) as total_students,
       sum(case when e.grade > acg.avg_grade then 1 else 0 end) as
num_greater_avg_grade
from courseclass cc
left join enrollment e on cc.clsno = e.clsno
left join avg_course_grade acg on cc.cno = acg.cno
group by cc.cno;

```

33 18:57:05 create view total_course_students(cno, total_students, num_greater_avg_grade) as select cc.cno, count(*) as total_students, ... 0 row(s) affected

```

(2) select c.cno, c.cname, c.dept
from courses c
join courseclass cc on c.cno = cc.cno
join enrollment e on cc.clsno = e.clsno
join avg_course_grade acg on c.cno = acg.cno
where e.grade is not null
group by c.cno, c.cname, c.dept, acg.total_students
having sum(case when e.grade < acg.avg_grade then 1 else 0 end) * 2 >
acg.total_students
order by c.dept asc, c.cno asc;

```

```

292 • select c.cno, c.cname, c.dept
293 from courses c
294 join courseclass cc on c.cno = cc.cno
295 join enrollment e on cc.clsno = e.clsno
296 join avg_course_grade acg on c.cno = acg.cno
297 where e.grade is not null
298 group by c.cno, c.cname, c.dept, acg.total_students
299 having sum(case when e.grade < acg.avg_grade then 1 else 0 end) * 2 > acg.total_students
300 order by c.dept asc, c.cno asc;
301

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	cno	cname	dept
▶	2205	数据结构	计算机
	1302	人工智能导论	人工智能
	2305	机器学习	人工智能

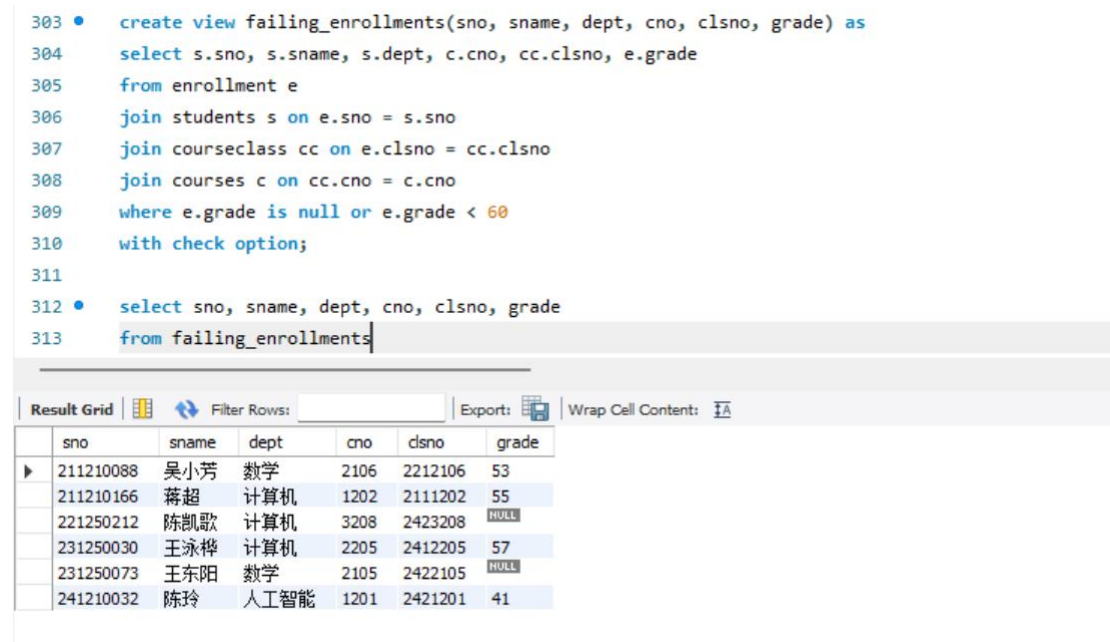
3.

(1)

```
create view failing_enrollments(sno, sname, dept, cno, clsno, grade) as
select s.sno, s.sname, s.dept, c.cno, cc.clsno, e.grade
from enrollment e
join students s on e.sno = s.sno
join courseclass cc on e.clsno = cc.clsno
join courses c on cc.cno = c.cno
where e.grade is not null or e.grade < 60;
with check option;
```

(2)

```
select sno, sname, dept, cno, clsno, grade
from failing_enrollments;
```



```
303 • create view failing_enrollments(sno, sname, dept, cno, clsno, grade) as
304 select s.sno, s.sname, s.dept, c.cno, cc.clsno, e.grade
305 from enrollment e
306 join students s on e.sno = s.sno
307 join courseclass cc on e.clsno = cc.clsno
308 join courses c on cc.cno = c.cno
309 where e.grade is not null or e.grade < 60
310 with check option;
311
312 • select sno, sname, dept, cno, clsno, grade
313 from failing_enrollments
```

	sno	sname	dept	cno	clsno	grade
▶	211210088	吴小芳	数学	2106	2212106	53
	211210166	蒋超	计算机	1202	2111202	55
	221250212	陈凯歌	计算机	3208	2423208	NULL
	231250030	王泳桦	计算机	2205	2412205	57
	231250073	王东阳	数学	2105	2422105	NULL
	241210032	陈玲	人工智能	1201	2421201	41

(3)

```
set autocommit=0;
```

(4) (5)

```
select *
from failing_enrollments
where grade is not null and grade < 60 limit 1;
```

```
update failing_enrollments
set grade = 60
where sno = '211210088' and clsno = '2212106';
```

```
ROLLBACK;
```

```

318 • select *
319   from failing_enrollments
320  where grade is not null and grade < 60 limit 1;
321
322 • update failing_enrollments
323   set grade = 60
324  where sno = '211210088' and clsno = '2212106';
325
326 • ROLLBACK;
327

```

sno	sname	dept	cno	clsno	grade
211210088	吴小芳	数学	2106	2212106	53

Output

#	Time	Action	Message	Duration / Fetch
52	19:14:19	update failing_enrollments set grade = 60 where sno = '20210001' and clsno = 'C001-01'	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.000 sec
53	19:14:28	select * from failing_enrollments where grade is not null and grade < 60 limit 1	1 row(s) returned	0.000 sec / 0.000 sec
54	19:14:47	update failing_enrollments set grade = 60 where sno = '211210088' and clsno = '2212106'	Error Code: 1369 CHECK OPTION failed 'myschool26.failing_enrollments'	0.000 sec
55	19:14:51	select * from failing_enrollments where grade is not null and grade < 60 limit 1	1 row(s) returned	0.000 sec / 0.000 sec
56	19:14:51	update failing_enrollments set grade = 60 where sno = '211210088' and clsno = '2212106'	Error Code: 1369 CHECK OPTION failed 'myschool26.failing_enrollments'	0.000 sec

(6) (7) (8)

update enrollment e

set grade = 70

where sno = '211210088' and clsno = '2212106';

select sno, sname, dept, cno, clsno, grade

from failing_enrollments;

ROLLBACK;

```

329 • update enrollment e
330   set grade = 70
331  where sno = '211210088' and clsno = '2212106';
332 • select sno, sname, dept, cno, clsno, grade
333   from failing_enrollments;
334
335 • ROLLBACK;

```

sno	sname	dept	cno	clsno	grade
211210166	蒋超	计算机	1202	2111202	55
221250212	陈凯歌	计算机	3208	2423208	NULL
231250030	王永桦	计算机	2205	2412205	57
231250073	王东阳	数学	2105	2422105	NULL
241210032	陈玲	人工智能	1201	2421201	41

Output

#	Time	Action	Message
55	19:14:51	select * from failing_enrollments where grade is not null and grade < 60 limit 1	1 row(s) returned
56	19:14:51	update failing_enrollments set grade = 60 where sno = '211210088' and clsno = '2212106'	Error Code: 1369 CHECK OPTION failed 'myschool26.failing_enrollments'
57	19:17:09	update enrollment e set grade = 70 where sno = '211210088' and clsno = '2212106'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
58	19:17:09	select sno, sname, dept, cno, clsno, grade from failing_enrollments LIMIT 0, 1000	5 row(s) returned
59	19:17:09	ROLLBACK	0 row(s) affected

(9)

update enrollment e

set grade = 50

where sno = '211210088' and clsno = '2212106';

60	19:19:23	update enrollment e set grade = 50 where sno = '211210088' and clsno = '2212106'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
----	----------	--	--

(10)

update enrollment

set grade = null

where grade is not null

and grade < 60

and clsno in (

select cc.clsno

from courseclass cc

join courses c on cc.cno = c.cno

where c.cname = '数据结构'

);

61 19:19:49 update enrollment set grade = null where grade is not null and grade < 60 and clsno in (select cc.clsno from courseclas... 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

(11) (12)

```
353 • select sno, sname, dept, cno, clsno, grade
354 • from failing_enrollments;
355
356 • ROLLBACK;
357
358 • ##### (实验2) 脚本结束 #####
359
```

sno	sname	dept	cno	clsno	grade
211210088	吴小芳	数学	2106	2212106	50
211210166	蒋超	计算机	1202	2111202	55
221250212	陈凯歌	计算机	3208	2423208	NULL
231250030	王泳桦	计算机	2205	2412205	NULL
231250073	王东阳	数学	2105	2422105	NULL
241210032	陈玲	人工智能	1201	2421201	41

#	Time	Action	Message
59	19:17:09	ROLLBACK	0 row(s) affected
60	19:19:23	update enrollment e set grade = 50 where sno = '211210088' and clsno = '2212106'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
61	19:19:49	update enrollment set grade = null where grade is not null and grade < 60 and clsno in (select cc.clsno from courseclas...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
62	19:20:17	select sno, sname, dept, cno, clsno, grade from failing_enrollments LIMIT 0, 1000	6 row(s) returned
63	19:20:17	ROLLBACK	0 row(s) affected

4.

drop view if exists avg_course_grade;

drop view if exists total_course_students;

drop view if exists failing_enrollments;

358 •	drop view if exists avg_course_grade;
359 •	drop view if exists total_course_students;
360 •	drop view if exists failing_enrollments;
361	
362	##### (实验2) 脚本结束 #####
363	

#	Time	Action	Message
62	19:20:17	select sno, sname, dept, cno, clsno, grade from failing_enrollments LIMIT 0, 1000	6 row(s) returned
63	19:20:17	ROLLBACK	0 row(s) affected
64	19:21:22	drop view if exists avg_course_grade	0 row(s) affected
65	19:21:22	drop view if exists total_course_students	0 row(s) affected
66	19:21:22	drop view if exists failing_enrollments	0 row(s) affected

实验中遇到的困难及解决办法

[详细说明你认为本次实验中比较困难的地方，也可以对实验设计提出建议]

- 一些不熟悉的函数需要搜索用法，比如字符串函数 `left`
- 没有弄清楚 `where` 和 `having` 的用法
- 对视图的操作不熟练
- `case when` 子句不熟练

参考文献及致谢

[如果你参考了任何书籍、网页，或与他人进行了讨论，请在此注明]

<https://claude.ai/chat> 询问字符串函数 `left` 的用法，以及 `'_'` 和 `'%'` 的区别